

SEMICONDUCTOR TRADE ANALYSIS

U.S. Semiconductor Imports — Full EDA & Findings | 2025

Data Source & Project Overview

- **Data Source:** UN Comtrade Plus
- **Objective:** Uncover the landscape of U.S. semiconductor imports in 2025, identifying key trade partners, market dynamics, and emerging trends. The primary goal was to demonstrate Malaysia's dominance in semiconductor exports to the United States while revealing how other markets — particularly Brazil, India, and Israel — are positioning themselves in this critical technology sector.

This analysis provides actionable insights into America's semiconductor supply chain dependencies, mapping not just who supplies the U.S., but how these relationships are evolving month by month — supporting supply chain risk assessment, strategic sourcing decisions, trade policy analysis, and market opportunity identification.

Phase 1 — Data Preparation & Initial Exploration

Objective

Load the cleaned dataset, perform essential data quality checks, ensure correct data types, and create derived fields for downstream analysis.

Key Steps

- Imported essential libraries: pandas for data manipulation, matplotlib and seaborn for visualization, scipy for statistical testing.
- Performed data type conversions for critical columns: cifvalue, fobvalue, netWgt, qty, altQty, refMonth, refYear.
- Created calculated field: value_per_unit to understand unit economics.
- Filtered out zero-value transactions to ensure meaningful analysis.

Initial Data Quality Assessment

- Used .head(), .info(), .dtypes, .shape(), and .describe() methods to validate structure.
- Ensured numeric columns were properly typed.
- Handled division-by-zero scenarios in unit value calculations.

Phase 2 — Distribution Analysis

Chart 1: Histogram of CIF Values (Linear Scale)

Purpose: Detect outliers and understand the natural distribution of import values. A 30-bin histogram with count annotations revealed an extremely right-skewed distribution, with the vast majority of transactions concentrated at lower values. This immediately signaled the presence of significant outliers — likely Malaysia's massive import volumes overshadowing other countries.

Chart 2: Histogram of CIF Values (Logarithmic Scale)

Purpose: Confirm and better visualize outliers by compressing the value range. The log scale confirmed a distinct separation between regular import transactions and extraordinarily high-value shipments. This validated the decision to create separate analyses with and without Malaysia.

Key Design Choice

- **Dual-histogram approach:** Using both linear and logarithmic scales is best practice in exploratory analysis when dealing with financial data that spans multiple orders of magnitude.

Phase 3 — Statistical Validation (Chi-Square Analysis)

Objective

Test whether there is a statistically significant relationship between exporter countries and import value categories.

Methodology

- Divided CIF values into tertiles (Low, Medium, High) using `pd.qcut()`.
- Created a contingency table using `pd.crosstab()`.
- Performed Chi-Square test of independence.
- Visualized with an annotated heatmap (blue color scheme).

Statistical Results

Metric	Value
Chi-Square Statistic	105.51
P-value	3.18e-15 (highly significant)
Degrees of Freedom	16

Key Design Choice

- **`pd.qcut()` vs `pd.cut()`:** `qcut()` creates equal-sized groups rather than equal-width bins, preventing empty cells that would invalidate the chi-square test.

The extremely low p-value (essentially zero) confirms that export volume categories are NOT independent of country of origin — certain countries consistently ship in certain value brackets. Malaysia dominates the 'High' category while emerging markets show different patterns.

Phase 4 — Country-Level Comparative Analysis

Chart 4: Distribution of CIF Value by Exporter Country (With Malaysia)

Box plots showing median, quartiles, and outliers for each exporter country. Malaysia's box plot was so compressed at the top of the scale that comparing other countries was nearly impossible — this chart effectively demonstrated why a separate 'without Malaysia' analysis was necessary.

Chart 5: Distribution of CIF Value by Exporter Country (Excluding Malaysia)

Filtering out Malaysian data (`reporterISO != 'MYS'`) revealed the competitive landscape among secondary suppliers. Germany emerged clearly as the second-largest supplier, Japan holds the third position with consistent volume, and Brazil, India, and Israel showed promising activity levels.

Key Design Choice

- **Discarded approach:** Violin plots were initially considered but did not provide clearer insights given the extreme outlier situation. Box plots were retained.

Phase 5 — Temporal Trends

Chart 6: CIF Value Over Time by Exporter Country

Monthly import trends throughout 2025, with month numbers converted to abbreviated names. Each line is annotated with its first value, maximum value, and last value. Malaysia shows consistent dominance with minimal fluctuation; seasonal patterns are visible in secondary suppliers.

Chart 7: CIF Value Trend Over Time (Excluding Malaysia)

Germany maintains steady, reliable export volumes. Japan shows slight seasonal variation. Emerging markets (Brazil, India, Israel) display more volatility — typical of growing trade relationships.

Chart 8: CIF Value Over Time — Malaysia vs Japan vs Hong Kong

Direct comparison of the top three markets with different characteristics. Malaysia is dominant and stable; Japan is a consistent mid-level supplier; Hong Kong exhibits fluctuations suggesting it may serve as a transshipment point for Chinese semiconductors.

Key Design Choice

- **Annotation strategy:** 45-degree rotated annotations prevent label overlap while maintaining readability across all trend lines.

Phase 6 — Logistics & Transportation Analysis

Chart 9: CIF Value by Exporter Country and Mode of Transport

Air freight dominates for high-value, time-sensitive shipments (particularly from Asia). Sea freight is preferred for bulk shipments from Malaysia. Mixed modal transport patterns reveal supply chain sophistication levels by country.

Chart 10: CIF Value by Country and Mode of Transport (Excluding Malaysia)

Without Malaysia's volume, Germany relies heavily on air freight (likely for specialized chips), while Brazil shows more balanced modal distribution — suggesting investment in logistics infrastructure.

Key Design Choice

- Null values and the 'TOTAL MOT' category were filtered out to ensure the chart reflected actual modal breakdowns.

Phase 7 — Monthly Market Share Analysis

Charts 11–22: Contribution to Total CIF Imports (Jan–Dec 2025)

Individual donut charts for each month of 2025, with countries below 2% grouped as 'Others'. Malaysia consistently captures 55–65% of monthly imports — a level of dominance that barely budges month to month. Germany fluctuates between 8–12%; Japan holds steady at 6–9%.

Charts 23–33: Contribution to Total CIF Imports (Excluding Malaysia, Jan–Nov 2025)

Germany becomes the clear leader (30–35% share without Malaysia), Japan takes second position (20–25%), and Brazil, India, and Israel each capture 5–10% individually. Competition among secondary suppliers is genuine and dynamic.

Key Design Choice

- **2% threshold:** Countries below this share are grouped as 'Others' to maintain chart readability while preserving accurate total percentages.

Phase 8 — Excel Analysis: CIF vs FOB Gap

Objective

Add an analytical layer in Excel to quantify the cost of insurance and freight as a percentage of goods value.

Formula

```
=IF(OR([@cifvalue]=0, [@fobvalue]=0), "", [@cifvalue]-[@fobvalue])
```

Key Findings

- **Malaysia:** Relatively low CIF-FOB gap despite high volume — likely due to established, efficient shipping routes.
- **Germany:** Higher gap as expected given the distance — European shipments carry more freight and insurance cost.
- **Brazil:** Competitive gaps despite being an emerging player, suggesting investment in logistics infrastructure.

Pivot Table: CIF vs FOB Values by Country

Country	Records	Sum of CIF Value	Sum of FOB Value	Sum of CIF-FOB Gap
Brazil	36	82,453,140	82,105,292	347,848
Canada	42	0	273,366,412.6	0
China, Hong Kong SAR	10	485,305,779.8	0	0
Germany	158	648,216,101.6	0	0
India	8	84,702,706.52	0	0
Israel	10	106,168,000	0	0
Japan	11	447,470,099.1	0	0
Malaysia	44	9,619,905,202	825,590,670	0
Mexico	43	0	825,590,670	0
Netherlands	8	71,039,227.73	0	0
Singapore	3	31,643,615.21	0	0
Grand Total	373	11,576,903,872	1,181,064,375	347,848

Phase 9 — Excel Dashboard

Purpose

Validate Python analysis findings and provide granular drill-down capability for each country.

Features

- Interactive filters to toggle between 'With Malaysia' and 'Without Malaysia' views.
- Country-specific breakdowns: number of shipments, total value, average value.
- Monthly trend sparklines for quick pattern recognition.
- Transport mode distribution by country.

Validation Outcome

Every insight from Python was confirmed in Excel. No discrepancies found — the data tells a consistent story across platforms.

Phase 10 — Power BI Dashboard

Purpose

Create a real-time interactive experience for exploring the dataset with maximum comfort and visual appeal.

Dashboard Components

- **KPI Cards:** Total import value, number of countries, average shipment value, year-over-year comparisons.
- **Interactive Map:** Geographic visualization of exporter countries with bubble sizes representing volume.
- **Dynamic Filters:** Country, month, transport mode, value ranges.
- **Drill-Through Pages:** Detailed country profiles accessible by clicking on any country in charts.

Key Design Choices

- **Custom color palette:** Aligned with U.S. flag colors for patriotic stakeholder appeal.
- **Page structure:** Malaysia-included view on page 1, Malaysia-excluded view on page 2 — mirroring the analytical journey.
- **Power BI vs Excel:** Power BI offers superior interactivity, better handling of large datasets, and more sophisticated DAX calculations for complex metrics.

Corrected Steps & Discarded Analyses

- **Average Transaction Value KPI:** Dropped due to extreme outlier distortion. Replaced with median transaction value and quartile ranges.
- **Heatmap of All Countries x All Months:** Dropped due to visual clutter. Replaced with separate time-series line charts.
- **Correlation Matrix of All Numeric Variables:** Dropped — CIF and FOB are naturally highly correlated. Replaced with the CIF-FOB gap as a derived metric.
- **CIF vs FOB Plot:** Dropped — only Brazil and Mexico had available data. Replaced with a pivot table using the calculated `cif_fob_gap` column.
- **Network Graph of Trade Relationships:** Dropped — bilateral U.S. import data only; network analysis requires multi-country flows to be meaningful.

Key Findings & Conclusions

Finding #1 — Malaysia's Overwhelming Dominance

Malaysia accounts for approximately 55–65% of all U.S. semiconductor imports by value throughout 2025. The U.S. has a significant dependency on Malaysian semiconductor supply chains — both a strength (reliable, high-volume partner) and a risk (single point of failure). Malaysia's position is unlikely to be challenged in the short term given established infrastructure.

Finding #2 — Germany is the Undisputed Second Supplier

When Malaysia is excluded, Germany captures 30–35% of the remaining market. European semiconductor technology — particularly specialized chips — holds a strong position in U.S. supply chains, representing a Western alternative to Asian supply chains.

Finding #3 — Japan Holds Steady at #3

Japan consistently maintains 6–9% market share (or 20–25% excluding Malaysia). Despite South Korea's massive semiconductor industry, Japan remains a more significant U.S. supplier in this commodity category — South Korea did not rank in the top 8 suppliers.

Finding #4 — Emerging Markets on the Rise

Brazil, India, and Israel each capture 5–10% of the non-Malaysia market. These three countries represent genuine growth opportunities and could be strategically developed to reduce Malaysian dependency.

Finding #5 — The Hong Kong–China Dynamic

Hong Kong exports more semiconductors to the U.S. than Shanghai in this specific trade data, continuing to serve as a critical gateway for Chinese semiconductor trade despite geopolitical tensions.

Finding #6 — The Canadian and Mexican Surprise

Despite USMCA partnership, Canada and Mexico barely registered in this dataset. Semiconductor components may flow through different customs classifications, or these countries may be net importers rather than exporters.

Recommendations for Stakeholders

- Monitor Malaysia concentration risk — develop contingency plans for Malaysian supply disruptions.
- Strengthen German and Japanese partnerships — these are reliable Western/allied alternatives.
- Invest in Brazil, India, and Israel relationships — these emerging markets offer diversification potential.
- Investigate Hong Kong routing patterns — understand what is actually Chinese-origin vs. Hong Kong-origin.
- Explore South Korean direct trade — why is this relationship not visible in this data?
- Re-examine USMCA semiconductor provisions — Canadian and Mexican participation is surprisingly low.

Technical Appendix

Key Libraries & Methods

Category	Tool / Method	Purpose
Data Loading	pd.read_csv()	Import official UN Comtrade data

Category	Tool / Method	Purpose
Type Conversion	pd.to_numeric()	Convert string values to numeric with error handling
Statistical Testing	pd.qcut()	Create tertile categories for Chi-Square testing
Statistical Testing	chi2_contingency()	Perform independence testing
Statistical Testing	pd.crosstab()	Build contingency tables
Visualization	matplotlib.pyplot	Base plotting functionality
Visualization	seaborn	Statistical visualization layer
Temporal Processing	calendar.month_abbr[]	Convert month numbers to readable names
Excel Formula	CIF-FOB gap formula	Zero-value handling with IF/OR logic
Power BI	DAX Measures	Market share %, YoY growth, running totals

Document Metadata

- **Prepared by:** Aaron E. Obando
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- **Data Source:** [UN Comtrade Plus \(Official\)](#)
- **Analysis Tools:** Python (pandas, seaborn, matplotlib, scipy), Microsoft Excel, Power BI